Adnan Hasnat

Software Engineer

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Career profile

Hello! I am a 2025 advanced computer science R&D and science double degree graduate from the Australian National University, specialising in computer systems and architecture. My main interests are computer microarchitecture and memory/storage systems, and I have experience in working on various low-level systems and frameworks through my research projects. I am looking to develop and contribute my skills as a software engineer, performance engineer and systems developer.

Skills

Programming languages learnt: Python, Java, C, C++, BASH
Environments used: Linux, remote servers

Transferrable skills: problem solving, liaising with various stakeholders, research and navigating difficult open problems including planning and design of research activities, communication and teamwork

Education

Bachelor of Advanced Computing R&D and Bachelor of Science (maths major) double degree
Australian National University
2021-2025
- University Medal 2025 winner
- Erin Brent Prize 2025 winner (for highest Honours mark in cohort)
- Perfect GPA of 7.0
- Gained experience with software engineering through courses (COMP2100 Software Construction, COMP2120 Software Engineering), which included building projects in Java
- Specialised in systems and architecture, which included building low level software in Java/C and scripting frameworks in Python and BASH. Key courses included COMP3770 Compiler Construction, COMP2310 Systems Networks and Concurrency, COMP4300 Parallel Systems and COMP3770 Microarchitecture.

Software development projects

SPIRIT: Scalable Persistent In-Memory Real-Time Search Engine
2022-2024
- Complex and optimised novel search engine designed to work with Intel Persistent Memory, written in C
- Gained experience building a system to be tolerant to failures and crashes, including transactions and logging
- Featured in Transactions on Architecture and Code Optimisation (TACO) 2025, and High Performance Edge and Cloud Computing (HiPEAC) 2025.
- Paper, conference talk and source code available.

Memory Level Parallelism extension for Sniper CPU Simulator
2024
- Contributed to an extension on an existing large C++ codebase, working across C++, Python and BASH
- Gained experience analysing complex performance data and full-stack performance optimisation
- Featured in Computing Architecture Letters (CAL) 2025
- Paper and source code available.

Miscellaneous additional projects (see links for source code)
- Java implementation of board games (Cublino, Glinski Hexagonal Chess)
- Java implementation of an Android app
- ARM-M v7 Assembly implementation of a light show for the BBC Microbit
- Java implementation of a compiler for a custom language
- C implementation of a memory allocator and HTTP proxy

Experience

Casual Sessional Academic (Tutor)
Australian National University
2023-2025
- Contributed to the design of software development assignments in C, including testing harnesses, specifications and example implementations. Examples of projects include a memory allocator and a distributed database.
- Gained teamwork and collaboration skills, liaising between the tutor team and students

Volunteering and extracurricular activities:
Common Room Officer for the Computer Science Students Association – 2025
Treasurer for the Improv Society – 2022-2024
Course Representative – 2021-2022